

Federated Learning of Deep Networks using Model Averaging

In this paper, the authors explore a privacy preserving, distributed approach to training LLMs that proves robust to non IID data distributions that naturally arise, parameter averaging over updates from multiple clients produces surprisingly good results, like decreasing the communication needed to train an LSTM language model by two orders of magnitude. The authors investigate a learning technique, termed Federated Learning, that allows users to collectively reap benefits of the distributed state of smartphones and the data available through their powerful sensors.

In federated Learning, the learning task is solved by a loose federation of participating devices (aka clients) which are coordinated by a central server. The authors introduce federated averaging algorithm which combines local SGD training on each client with communication rounds where the central server performs model averaging. Their experiments show that this technique is robust to unbalanced and non-IID data distributions. Federated learning has several use cases, however, ideally the problems should have these properties; training on real-world data from mobile devices should provide a distinct advantage over training on proxy data that is generally available in the data-center, this data is privacy sensitive or large in size, and for supervised tasks, labels on the data can be inferred naturally from a user's interaction with their device.

Federated Learning: Collaborative Machine Learning without Centralized Training Data

The article gives a brief overview of federated learning. Federated Learning enables mobile phones to collaboratively learn a shared prediction model while keeping all the training data on device, decoupling the ability to do machine learning from the need to store the data in the cloud. A given device downloads the current model, improves it by learning from data on the phone and then summarizes the changes in a small update, which is then sent to the cloud. On the cloud, this update is averaged with other updates to improve the shared model.

Federated Learning allows for smarter models, lower latency, and less power consumption, all while ensuring privacy. Another benefit of this approach is that the updated model on the phone can be used immediately. In traditional machine learning setting, algorithms like stochastic gradient descent require datasets to be partitioned homogeneously across servers, with low latency, high throughput connections. In federated learning, this data is distributed across millions of devices with varying power and throughput. These bandwidth and latency limitations motivate the Federated Averaging algorithm, which can train deep networks using 10-100x less communication compared to a naively federated version of SGD. The key idea is to use the powerful processors in modern mobile devices to compute higher quality updates than simple gradient steps. Since it takes fewer iterations of high-quality updates to produce a good model, training can use much less communication.

To reduce the upload communication cost, the updates are compressed using random rotations and quantization. On device training uses a miniature version of tensorflow. Careful scheduling ensures training happens only when the device is idle, plugged in, and on a free wireless connection, so there is no impact on the phone's performance. The system then needs to communicate and aggregate the model updates in a secure, efficient, scalable, and fault-tolerant way.

Federated learning does not require storing of the user data in the cloud, but the engineers at Google research have also developed a secure aggregation protocol that uses cryptographic techniques so a coordinating server can only decrypt the average update if 100s or 1000s of users have participated, and no individual phone's update can be inspected before averaging. Applying Federated Learning requires machine learning practitioners to adopt new tools and a new way of thinking: model development, training, and evaluation with no direct access to or labeling of raw data, with communication cost as a limiting factor.

This article was really interesting, especially since it is coming from Google. Google has historically been criticized for storing user data for various purposes and with machine learning on the rise, there have been increasing privacy concerns. With federated learning, the data from an individual device cannot be inspected and average values are utilized, which makes it much more privacy preserving and opens avenues for future research.